# Lab 01

## Atlantic environmental conditions

Teaching Week 01 | Individual work. Submit in R or Python.

> Integrated into Class B on September 10, 2026. Individual work in R or Python. Due September 12, 2026 at 23:59 Halifax time. The best 10 of 12 labs count for 15% of the course grade. Generative AI output is prohibited on integrated labs.

Use local R or Python, Posit Cloud, Kaggle Notebooks, or Google Colab (Python). Keep assessed work and cloud notebooks private until the deadline and the instructor opens sharing. After that release, publish and present the same submitted commit as part of your data science portfolio. Follow the submission and history-preservation instructions below.

> **Purpose**
>
> Describe an open Atlantic environmental dataset using the observations, summaries, and plots introduced in Classes A and B. Investigate unusual readings against the source documentation.

## Learning outcomes

After completing this lab, you should be able to

- distinguish observations, variables, identifiers, groups, and units;
- compute and interpret centre, spread, quantiles, and missingness;
- investigate potential outliers in their environmental context; and
- document an open dataset so another analyst can retrieve it.

## Dataset choice

Choose one of the verified open datasets below. Each option supports the assessment when your analytical table satisfies the data contract in the next section. The expected variables describe the published record and may differ in detail from the extract you retrieve; reconcile any difference in your report.

**Option A | Nova Scotia Inland Water Quality Dataset**

**Publisher.** Nova Scotia Fisheries and Aquaculture
**Official record.** https://data.novascotia.ca/d/qaf8-y5x2
**Record identifier.** `qaf8-y5x2`
**Licence and format.** Nova Scotia Open Government Licence; CSV
**Expected variables.** county; waterbody; station; timestamp; latitude; longitude; sensor depth; dissolved oxygen saturation; temperature; quality flags
**Use in this lab.** describe and compare inland-water conditions
**Why it fits.** Repeated sensor measurements support summaries, spread, and unusual-value investigation.

Filter to a manageable waterbody, station, and period before exporting CSV. Temperature, dissolved oxygen saturation, and measured sensor depth are candidate measurements. Availability depends on sensor type; inspect non-missing counts before selecting the extract. Keep the quality flags and acknowledge the Centre for Marine Applied Research (CMAR).

**Option B | Meteorological Observations**

**Publisher.** Environment and Climate Change Canada
**Official record.** https://open.canada.ca/data/en/dataset/493966f9-f683-4e56-8fa6-8799999c00bd
**Record identifier.** `493966f9-f683-4e56-8fa6-8799999c00bd`
**Licence and format.** Open Government Licence - Canada; CSV through the historical climate data service
**Expected variables.** station; date; temperature; precipitation; wind
**Use in this lab.** describe weather conditions across stations
**Why it fits.** Numeric observations and station groups support distributional comparisons.

Use Historical Climate Data to choose a station in Nova Scotia, New Brunswick, Prince Edward Island, or Newfoundland and Labrador. Export daily or hourly CSV for a period long enough to meet the contract. Check which temperature, precipitation, and wind measurements are available; record the station, interval, missing-value symbols, and units.

**Option C | Halifax County Wave Data**

**Publisher.** Nova Scotia Fisheries and Aquaculture
**Official record.** https://data.novascotia.ca/d/9gpk-gb2v
**Record identifier.** `9gpk-gb2v`
**Licence and format.** Nova Scotia Open Government Licence; CSV
**Expected variables.** waterbody; station; deployment; timestamp; wave height; wave period; wave direction; water speed; water direction; sensor depth; quality flags
**Use in this lab.** describe coastal wave conditions across deployments or periods
**Why it fits.** Continuous Atlantic measurements support summaries, spread, and unusual-value checks.

Filter to a station, deployment, and period before exporting CSV. Wave height, wave period, water speed, and measured depth are candidate measurements. Retain the gross-range quality flags and acknowledge CMAR. Direction fields are circular measurements and do not count toward the three arithmetic variables.

## Data contract

Choose one listed dataset and produce an observation-level analytical table.

- Retain at least 100 observations and three numeric variables.
- Use three physical measurements with enough observed values for meaningful summaries. Report each variable's non-missing count; an all-missing column does not satisfy the contract. Coordinates, identifiers, timestamps, and quality-flag codes do not count as the three measurements.
- Choose measurements suited to ordinary arithmetic summaries, such as temperature, oxygen saturation, depth, precipitation, wind speed, wave height, or wave period. Do not average compass directions as ordinary numbers.
- Identify the observational unit and any site, station, region, depth, or time grouping.
- Preserve units, keep stable row references, and distinguish missing values from measured zeroes.
- Document the official record URL, publisher, licence, retrieval steps, and access date.
- Compute centre, spread, quantiles, and missingness for defensible variables.
- Investigate unusual values without deleting them automatically.
- Keep identifiers out of numerical summaries.

**What the contract does not settle**

Meeting the contract establishes that the table is usable. It does not establish that your analytical choices or interpretation are sound.

## Selection checklist

Resolve every unchecked item before you begin the analysis.

- ☐ I chose one option from the dataset catalogue above.
- ☐ I recorded the official URL, publisher, licence, access date, and retrieval steps.
- ☐ I can state the observational unit in one sentence.
- ☐ I identified numeric variables, units, groups, identifiers, and missing-value codes.
- ☐ My table satisfies every rule in the data contract.
- ☐ I can compute and interpret centre, spread, quantiles, and unusual values.
- ☐ I will keep raw data separate from cleaned analytical data.
- ☐ I will cite the dataset in the report.

## Cumulative method

This lab carries forward the observational-unit, variable-type, and analytical-question distinctions introduced in Class A. Your report must show how those earlier decisions support the current analysis rather than repeating them as disconnected exercises.

## Work program

### 1. Establish provenance and purpose

Record the publisher, official URL, licence, access date, retrieval steps, observational unit, and one focused descriptive question.

### 2. Audit the table

Build a variable table with type, unit, missingness, identifier status, and analytical role. Explain exclusions and document any conversion of missing-value codes. Advanced cleaning methods are introduced next week.

### 3. Describe the distributions

For each of the three measurements, report the non-missing count, missing count, mean, median, sample standard deviation, range, and first and third quartiles. Compare at least one measurement between two meaningful groups, such as two stations or two stated periods, and report each group's count. Use the Class B worked example as the starting pattern. Use the observed count for each summary, the $n - 1$ variance divisor, and type-7 quartiles (linear interpolation in pandas). Define the comparison groups before examining outliers and preserve their units.

### 4. Investigate unusual observations

Apply the type-7 boxplot rule and state whether its fences describe all observations or each comparison group separately. Use one context-based check, such as a published valid range, quality flag, or original measurement record. Trace several flagged cases back to source context and decide whether to retain, correct, or exclude each one. If no cases are flagged, report that result and inspect the most extreme observations instead. Do not create outliers to satisfy the task. A publisher's quality flag and the statistical boxplot flag answer different questions; compare them and explain any disagreement.

### 5. Communicate the baseline

Present one clear table and two plots with units, concise interpretations, and one limitation of the available observations. Repeated measurements at nearby times or locations are not independent replicates. Keep conclusions descriptive and limited to the observed places and period; this lab does not require a predictive model or significance test.

## Reproducible workflow

You may complete and submit the lab in either R or Python. Both paths are marked against the same evidence and rubric, and both must run end to end from one documented entry point in your portfolio repository.

Use the import pattern from the Class B practice. Replace the example column names with three physical measurements from your chosen extract. Keep its original download separately. Here `data/observations.csv` is the local copy read by your script; the path is relative to `labs/lab-01/`.

```r
data_path <- "data/observations.csv"
numeric_cols <- c("temperature_c", "oxygen_percent", "depth_m")
observations <- read.csv(data_path, stringsAsFactors = FALSE)
dir.create("results", showWarnings = FALSE)
```

**Listing 1.** Start of lab-01.R.

```python
from pathlib import Path
import pandas as pd

data_path = "data/observations.csv"
numeric_cols = ["temperature_c", "oxygen_percent", "depth_m"]
observations = pd.read_csv(data_path)
Path("results").mkdir(exist_ok=True)
```

**Listing 2.** Start of lab-01.py.

Continue with the summaries, comparison, screening, and plotting steps taught in Class B. Run from `labs/lab-01/` using `Rscript lab-01.R` or `python lab-01.py`, or restart and run all cells of your notebook. The small classroom dataset demonstrates the method; it cannot be submitted as the lab dataset. This lab uses no predictive target or random procedure.

Keep a separate, unchanged copy of the retrieved data and record every cleaning decision in code. Save the tables and figures cited by the report in the private work repository. These data and results are not course source. Publish that evidence in your public GitHub portfolio repository only after the deadline and the instructor's release. Link rather than redistribute raw data unless its licence and size permit it.

## Submission package

Keep the lab private before its submission deadline, including cloud notebooks and answer-revealing links. A branch in a public repository is not private. Give the instructor access to your private work repository. By 23:59 two calendar days after the integrated lab, submit the full commit SHA and its commit URL through the announced private submission channel. The immutable commit or release URL must resolve to that exact version of `labs/lab-XX/`. A branch URL or movable tag alone is not a fixed submission. Include

- a concise PDF report that answers the analytical question, together with the completed selection checklist;
- the complete R or Python source plus the generated tables, metrics, and figures referenced in the report;
- a README with the official data URL, retrieval instructions, exact execution command, and environment; and
- no redistributed source data unless its licence and size permit it.

After the deadline and when the instructor opens sharing, publish the same submitted commit and its history in your public GitHub portfolio repository and present that version in class. Preserve the submitted SHA when merging. Commit regularly as you work. Do not amend, rebase, squash, delete, replace, force-push, or backdate recorded assessment history. Keep submitted commits reachable and make corrections in new commits. The submission receipt and instructor-retained snapshot provide the independent assessment reference; Git author dates alone are not proof of timely submission.

## Assessment standard

The course uses one rubric across all integrated labs. The best ten of the twelve weekly lab results contribute 15% of the course grade.

**Table 1.** Integrated lab marking criteria.

| Criterion | Weight (%) |
|---|---|
| Dataset provenance and suitability | 15 |
| Reuse of earlier methods | 20 |
| Current-week analytical method | 35 |
| Interpretation and limitations | 20 |
| Reproducibility and communication | 10 |
| Total | 100 |

Academic integrity, private GitHub exceptions, private grading, and publication limits follow the course syllabus. Cite the dataset and every non-trivial external source; the submitted analysis must be your own work.